Synchronous startup phases

Validate topology; prepare files
compile each switch

Register cleanup hooks
create namespace; record; configure

Create veth pair
record pair; configure endpoints

Start BMv2
record handle; wait for readiness

Connect synchronous client
record client; load pipeline

Check IPv6 readiness; publish views
set running flag

exception

Startup exception
clean up; re-raise

Stored runtime references

Host namespaces

Veth pairs

BMv2 handles

Sync clients

Async clients attach
to running switch views

stop / astop request
wait for startup to settle

Shared cleanup pass

Reap user-spawned helpers

Submit async disconnects
to owning loops; wait on futures

Disconnect synchronous clients

Stop BMv2 processes

Destroy veth pairs

Remove host namespaces

Unregister hooks; clear runtime views
and running flag

Dashed horizontal arrows mark reference storage after helper return.

Cleanup errors are logged while later resources are attempted; pending async futures remain tracked.